

CENG218 – Labwork 7

Write a C++ program that implements the following class (FIFO Queue) for managing spacecrafts on space stations:

Queue
– arr*: int – front: int – rear: int – capacity: int – servedCount: int – name: string + Queue(int, string) + isFull(): bool + isEmpty(): bool + enqueue(int): void + dequeue(): void + display(): void + getServedCount(): int + getName(): string

arr* is representing the dynamic number of (capacity) spacecrafts, front and rear are representing queue's front and rear, servedCount is the number of spacecrafts that have visited the station (when the spacecraft takes off, this value is incremented (i.e. after dequeueing)), name is the name of the spacecraft, isFull(), isEmpty(), enqueue(), dequeue() are queue functions, Queue() parameterized constructor is for space station initialization (spacecraft capacity, space station name), and display function will display the current spacecrafts present at the space station.

After implementing this class, in the main function, create two space stations of this type and add number of static spacecrafts to each station, and take off different number of spacecrafts from them, finally compare which station is visited more and display its name on the screen. A sample output of the program can be observed below.

Sample Output:

```
// Adding spacecrafts with enqueue() function
Spacecraft 101 added to Station Elysium
Spacecraft 102 added to Station Elysium
Spacecraft 201 added to Station K-7
Spacecraft 202 added to Station K-7
Spacecraft 203 added to Station K-7
// Taking off spacecrafts with dequeue() function
// observe the queue behavior (FIFO)
Station Elysium docking spacecraft: 101
Station Elysium docking spacecraft: 102
Station K-7 docking spacecraft: 201
// Display the current situation of the space stations
Station Elysium queue is empty.
Station K-7 Queue: 202 203
// Make the comparison and print the result
Station Elysium was used more.
```